HEAT SHIELD

5 surprising facts

1.In 2010 , 4,190 workers lost work days due to work illness and 40 died .18 of those deaths were in construction industry

It is assumed that the number are higher , because many heat illnesses are never recognized or reported(NIOSH Ch.2, p 12)

2.Many workers were wrongly taught that, if you are still sweating , you cannot get heat stroke (NIOSH ,Executive summary , p,vi)

3.Accilimatization was not included by companies and was linked to most death cases(NIOSH p.12 and p.33)

4.Crop workers die from at about 20x the rate of all US workers (0.39 vs0.02 per 100,000)(NIOSH, p.12)

5. There is still no federal heat law .OSHA published a proposed rule on Aug 30, 2024. Hearing ran June 16-July 2,2025 and post hearning comments closed Oct 30, 2025.

the rule required that employers need to make a plan to evaluate and control heat hazards (OSHA rulemaking page)

Acclimatization

It is the body getting used to heat over several days in a row .It sweats sooner and more and the heart rate and body temperature stay lower (NIOSH, Glossary p xix, p.32) Your body needs 7 to 14 days to get used to heat, and about 50% longer if you are not fit(NIOSH Table 4-1, p. 34)

Solution

1.New workers , for the first day no more than 20 percent work post that one may allow more than 20 percent work per day

2.If an experienced worker then 50 percent of work for day 1 , 60 percent on day 2 80 on day 3 and 100 on day 4

3.A week or more away from heat will mean much of he adaptation is lost .it coms back in 2-3 days

4.Workers must replace the water they sweat out, or acclimatization slows down(NIOSH Table 4-1.p,34)

Some more interesting facts

1.Water -1 cup (8 oz) every 15-20 mins .Use sports drinks if sweating for more than 2 hours(NIOSH p.viii p.9)

2.Heat alert -forecast high of 95 degree Fahrenheit or more .or 90 degree Fahrenheit hotter than the days before(NIOSH p 10)

3.Heat stroke =emergency call 911 and cool the worker faster (NIOSH p.5 Table 4-3)

OSHA proposed rule -intial trigger 80 degree Fahrenheit heat index , high heat trigger 90 degree heat index

The problem is

New workers are put on full shifts in the heat with no slow start ,which is the gapmost linked to heat deaths .Small employers rarely fix it , because most states (including Illinois)have no heat rule and writing a plan means reading a regulation , a NIOSH document and a weather forecast at the same time(NIOSH ,p,12. Brief section 1.1)